Paper Outline

0. Abstract

1. Introduction

2. Problem Definition

Input

- Events
- Other live but less credible sources (such as social media)
- Cached information (for less latency)
- Constraints
    - latency budget
    - misinformation base rate (calculated by historic frequency)
    - FP/FN cost ratio

Output per LLM

- Real
- Fake
- Escalate
- Reasoning for human review

Decision

- Real
- Fake
- Escalate

Costs

- False positive cost
- False negative cost
- Latency cost

Different domains require different strategies.

3. Architecture

Standalone Systems

Single-Shot

- One LLM call
- Lowest latency
- higher false positives

- can be confused when event is complicated

Voting N

- N voters
- If Escalate rate > E: escalate
- Otherwise dominated answer is the final output
- Less latency than MoA

MoA

- Multiple roles:
  - believer
  - skeptic
  - judge
- Purpose: To show contradiction
- Longest latency
- Best results when paired with RAG

Routing

Requires historic data as arbitrary input

- Latency-first
  - Shortest possible: Single-Shot
  - Voting
- Cost first
  - MoA + RAG

4. Dataset

Domains

- Finance
- Healthcare
- Political

Analysis

- System qualities
- Base-rate analysis

Limitations

- RAG is simulated based on historic data in this research

5. Experiment Results

No-RAG comparison

Architecture	F1	Precision	Recall	FPR	Latency	Escalate
Single-Shot	0.645	0.780	0.733	0.333	3.9s	3%
Voting N=3	0.827	0.944	0.800	0.067	13.3s	23%
MoA	0.747	0.738	0.800	0.267	19.2s	10%

Analysis

- Voting has best overall result
- Voting is most likely to trigger escalation
- MoA has worst latency with generic results

RAG Impact

F1 scores:

RAG	Single-Shot	Voting N=3	MoA
OFF	**0.645**	0.827	0.747
ON	0.413	**0.857**	**0.917**

Analysis

- Single-Shot is worse when combined with RAG
- MoA receives a large improvement with RAG

Choices

- Single-Shot: ≤5s, tightest latency
- Voting: 11–16s, useful when slower decision allowed
- MoA+RAG: slow, better combined with RAG, useful for human review since it gives reasoning on different sides

6. Sensitivity Analysis

Base-rate

FP FN cost

- Change decision thresholds based on FP FN costs

Escalation

- Requires human review: labor costs
- FP FN rate of escalation

7. Conclusion

8. References

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